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CSE3016 – AWS SOLUTION ARCHITECT

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EXP.NO: 01	Launch an EC2 instance using the AWS Management Console and deploy a website using Apache middle ware
DATE: 10-11-25	

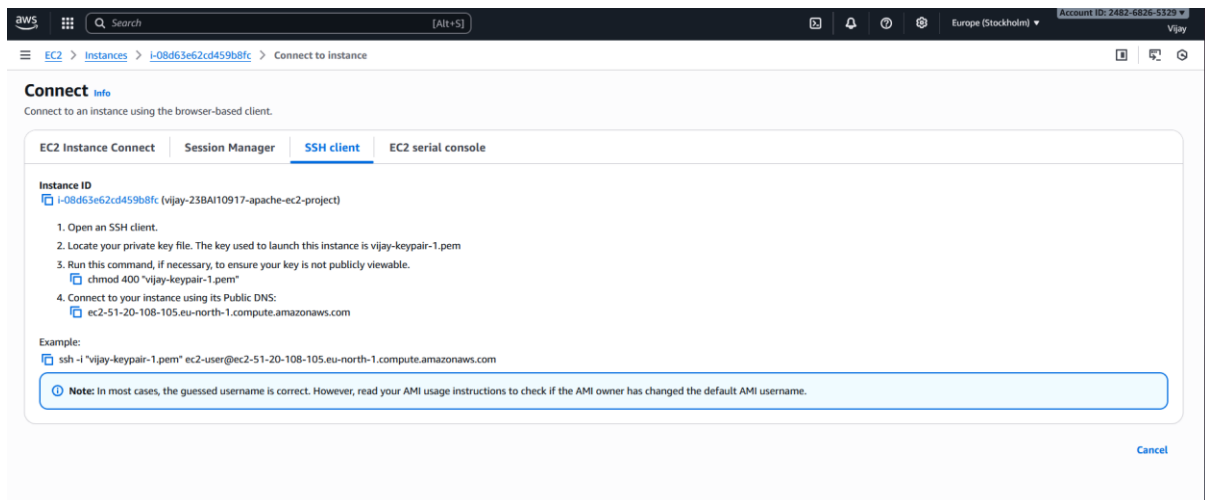
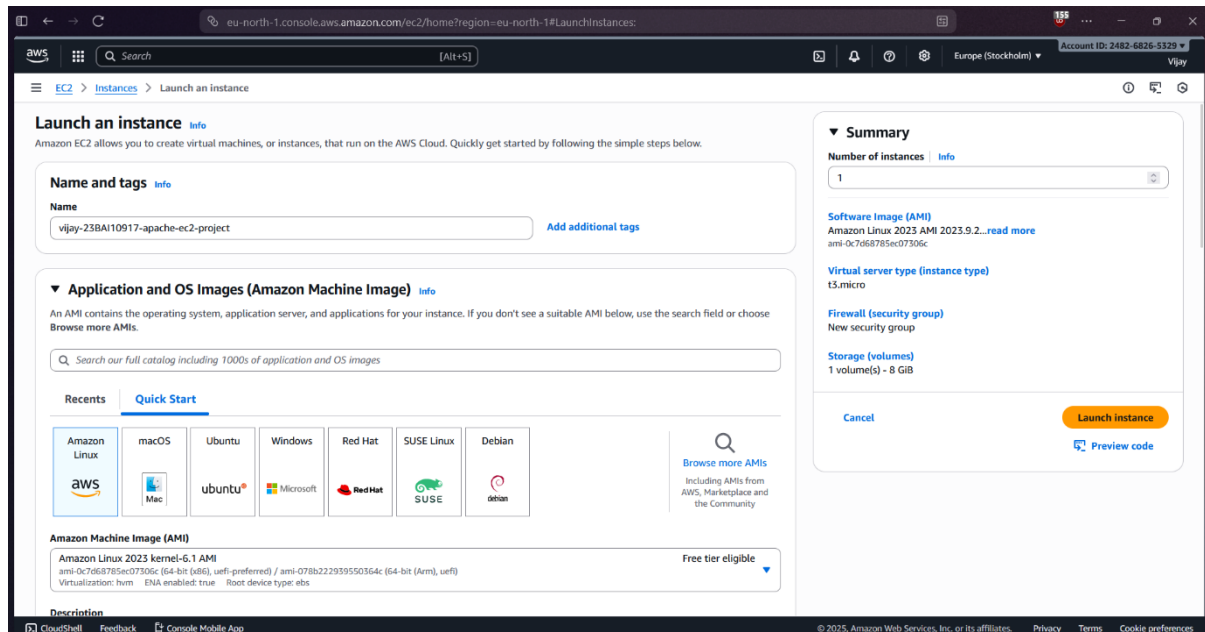
AIM

To Launch an EC2 instance using the AWS Management Console and to successfully deploy a website using Apache middle ware

PROCEDURE

1. Launch EC2 Instance: Use the AWS Console to launch a t2.micro instance with Amazon Linux 2023 and an appropriate Key Pair.
2. Configure Security Group: Ensure the instance's Security Group allows inbound traffic on Port 22 (SSH) and Port 80 (HTTP).
3. Connect via SSH: Use your downloaded .pem key file to SSH into the instance using the public IP address.
4. Install Apache: Run `sudo dnf update -y` then `sudo dnf install httpd -y` on the instance.
5. Start and Enable Service: Run `sudo systemctl start httpd` and `sudo systemctl enable httpd`.
6. Deploy Website: Navigate to `/var/www/html/` and create or upload your website files (e.g., `sudo nano index.html`).
7. Test: Open the instance's Public IPv4 Address in a web browser to view the deployed website.

IMPLEMENTATION SCREENSHOT




```

[root@ip-172-31-31-177 ec2-user]# yum install -y httpd.x86_64
Amazon Linux 2023 Kernel Livepatch repo 273 kB/s | 29 kB    00:00
Dependencies resolved.
=====
Package                Arch    Version                               Repository    Size
=====
Installing:
httpd                   x86_64  2.4.65-1.amzn2023.0.2               amazonlinux   47 k
Installing dependencies:
apr                     x86_64  1.7.5-1.amzn2023.0.4               amazonlinux   129 k
apr-util                x86_64  1.6.3-1.amzn2023.0.2               amazonlinux   97 k
apr-util-ldb            x86_64  1.6.3-1.amzn2023.0.2               amazonlinux   13 k
generic-logos-httpd    noarch  18.0.0-12.amzn2023.0.3             amazonlinux   19 k
httpd-core              x86_64  2.4.65-1.amzn2023.0.2               amazonlinux   1.4 M
httpd-filesystem        noarch  2.4.65-1.amzn2023.0.2               amazonlinux   13 k
httpd-tools             x86_64  2.4.65-1.amzn2023.0.2               amazonlinux   81 k
libbrotli               x86_64  1.0.9-4.amzn2023.0.2               amazonlinux   315 k
mailcap                 noarch  2.1.49-3.amzn2023.0.3               amazonlinux   33 k
Installing weak dependencies:
apr-util-openssl        x86_64  1.6.3-1.amzn2023.0.2               amazonlinux   15 k
mod_http2               x86_64  2.0.27-1.amzn2023.0.3               amazonlinux   166 k
mod_lua                 x86_64  2.4.65-1.amzn2023.0.2               amazonlinux   60 k

Transaction Summary
=====
Install 13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-ldb-1.6.3-1.amzn2023. 369 kB/s | 13 kB    00:00
(2/13): apr-1.7.5-1.amzn2023.0.4.x86_64 3.0 MB/s | 129 kB    00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x 2.1 MB/s | 97 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn20 737 kB/s | 15 kB     00:00
(5/13): httpd-2.4.65-1.amzn2023.0.2.x86 2.5 MB/s | 47 kB     00:00
(6/13): generic-logos-httpd-18.0.0-12.a 786 kB/s | 19 kB     00:00
(7/13): httpd-filesystem-2.4.65-1.amzn2 578 kB/s | 13 kB     00:00
(8/13): httpd-tools-2.4.65-1.amzn2023.0 2.9 MB/s | 81 kB     00:00
(9/13): httpd-core-2.4.65-1.amzn2023.0. 28 MB/s | 1.4 MB    00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2 9.9 MB/s | 315 kB    00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3. 1.3 MB/s | 33 kB     00:00
(12/13): mod_http2-2.0.27-1.amzn2023.0. 6.4 MB/s | 166 kB    00:00
(13/13): mod_lua-2.4.65-1.amzn2023.0.2. 2.8 MB/s | 60 kB     00:00
-----
Total                               14 MB/s | 2.4 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
  Installing     : apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64 2/13
  Installing     : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_ 3/13
  Installing     : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
  Installing     : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13

```

```

(13/13): mod_lua-2.4.65-1.amzn2023.0.2. 2.8 MB/s | 60 kB      00:00
-----
Total                               14 MB/s | 2.4 MB      00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
  Installing     : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 2/13
  Installing     : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
  Installing     : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
  Installing     : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13
  Installing     : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64 6/13
  Installing     : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7/13
Running scriptlet: httpd-filesystem-2.4.65-1.amzn2023.0.2.noa 8/13
  Installing     : httpd-filesystem-2.4.65-1.amzn2023.0.2.noa 8/13
  Installing     : httpd-core-2.4.65-1.amzn2023.0.2.x86_64 9/13
  Installing     : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 10/13
  Installing     : mod_lua-2.4.65-1.amzn2023.0.2.x86_64 11/13
  Installing     : generic-logos-httpd-18.0.0-12.amzn2023.0.3 12/13
  Installing     : httpd-2.4.65-1.amzn2023.0.2.x86_64 13/13
Running scriptlet: httpd-2.4.65-1.amzn2023.0.2.x86_64 13/13
  Verifying      : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
  Verifying      : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13
  Verifying      : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 3/13
  Verifying      : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
  Verifying      : generic-logos-httpd-18.0.0-12.amzn2023.0.3 5/13
  Verifying      : httpd-2.4.65-1.amzn2023.0.2.x86_64 6/13
  Verifying      : httpd-core-2.4.65-1.amzn2023.0.2.x86_64 7/13
  Verifying      : httpd-filesystem-2.4.65-1.amzn2023.0.2.noa 8/13
  Verifying      : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64 9/13
  Verifying      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
  Verifying      : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
  Verifying      : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13
  Verifying      : mod_lua-2.4.65-1.amzn2023.0.2.x86_64 13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64
apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.65-1.amzn2023.0.2.x86_64
httpd-core-2.4.65-1.amzn2023.0.2.x86_64
httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch
httpd-tools-2.4.65-1.amzn2023.0.2.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.65-1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-31-177 ec2-user]# |

```

```
Complete!
[root@ip-172-31-31-177 ec2-user]# systemctl start httpd.service
[root@ip-172-31-31-177 ec2-user]# systemctl enable start httpd.service
Failed to enable unit: Unit file start.service does not exist.
[root@ip-172-31-31-177 ec2-user]# systemctl enable httpd.service
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-31-177 ec2-user]#
```

The screenshot displays the AWS Management Console interface for the 'Instances' page. The top navigation bar shows the AWS logo, a search bar, and the account ID '2482-6826-5329'. The left sidebar contains navigation links for 'EC2', 'Instances', 'Dashboard', 'AWS Global View', 'Events', 'Instances' (expanded), 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Capacity Manager', 'Images' (expanded), 'AMIs', 'AMI Catalog', and 'Elastic Block Store' (expanded), 'Volumes', 'Snapshots', and 'Lifecycle Manager'.

The main content area is titled 'Instances (1/1) Info'. It features a table with columns: Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4. A single instance is listed: 'vijay-23BAI10...' with Instance ID 'i-08d63e62cd459b8fc', state 'Running', type 't3.micro', and availability zone 'eu-north-1a'. The 'Status check' column shows 'Initializing'.

Below the table, the details for instance 'i-08d63e62cd459b8fc (vijay-23BAI10917-apache-ec2-project)' are shown. The 'Instance summary' section includes the Instance ID, IP address, and Hostname type. The 'Public IPv4 address' is '51.20.108.105'. The 'Instance state' is 'Running'. The 'Private IP DNS name (IPv4 only)' is 'ip-172-31-31-177.eu-north-1.compute.internal'. The 'Public DNS' is 'ec2-51-20-108-105.eu-north-1.compute.amazonaws.com'.

Test Page

This page is used to test the proper operation of the Apache HTTP server after it has been installed. If you can read this page, it means that the Apache HTTP server installed at this site is working properly.

If you are a member of the general public:

The fact that you are seeing this page indicates that the website you just visited is either experiencing problems, or is undergoing routine maintenance.

If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.

For example, if you experienced problems while visiting `www.example.com`, you should send e-mail to `"webmaster@example.com"`.

If you are the website administrator:

You may now add content to the directory `/var/www/html/`. Note that until you do so, people visiting your website will see this page, and not your content. To prevent this page from ever being used, follow the instructions in the file `/etc/httpd/conf.d/welcome.conf`.

You are free to use the image below on web sites powered by the Apache HTTP Server:



RESULT

Thus, EC2 instance has been launched using the AWS Management Console and a website is deployed using the Apache Middle ware

EXP.NO: 02	Deploy a static website in S3 and manage the website using S3 life cycle management
DATE: 11-11-25	

AIM

To deploy a static website in S3 and manage the website using S3 Life Cycle Management, in the AWS Management Console

PROCEDURE

Step 1: Create and Configure the S3 Bucket

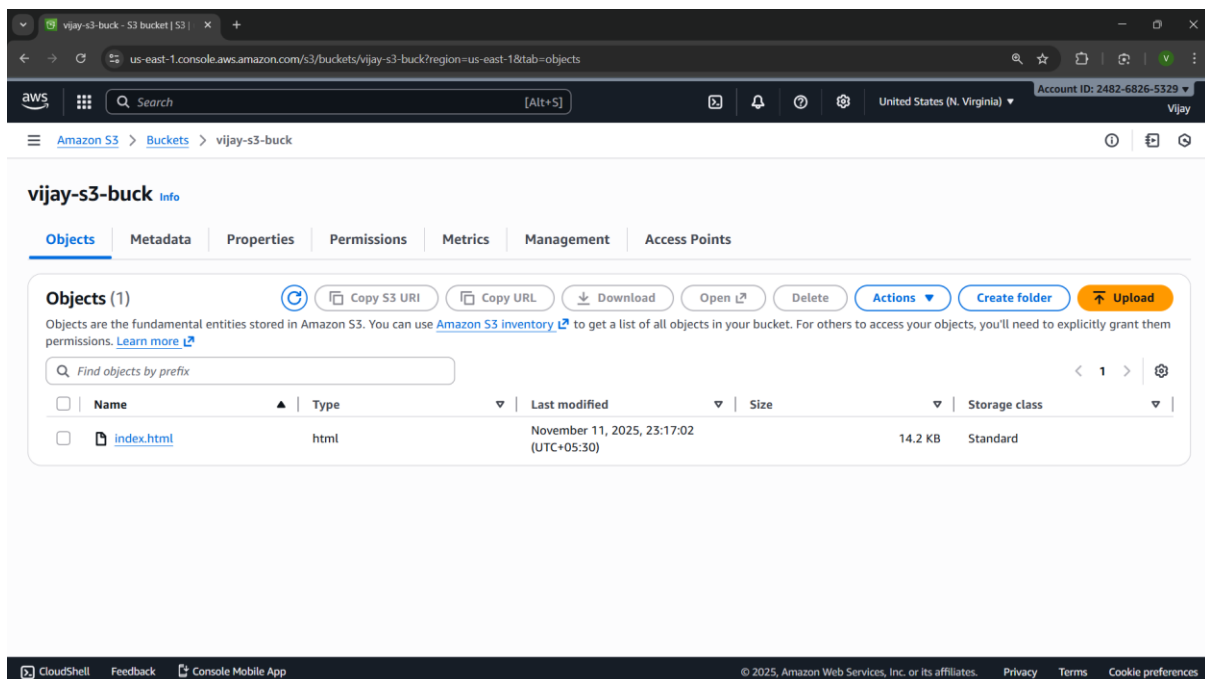
- **Create Bucket:** In the AWS S3 Console, click "Create bucket". Give it a globally unique name (e.g., my-static-site).
- **Public Access:** Uncheck "Block all public access" during creation (you'll set permissions next).
- **Enable Website Hosting:** Go to the bucket's Properties tab, find Static website hosting, click "Edit", select "Enable", and specify your Index document (e.g., index.html) and Error document (e.g., error.html). Save changes.

Step 2: Upload Files and Set Permissions

- **Upload Content:** Go to the Objects tab and Upload all your website files (HTML, CSS, images).
- **Apply Bucket Policy:** Go to the Permissions tab and click "Edit" under Bucket Policy. Paste a policy granting public s3:GetObject access to all resources in your bucket, replacing YOUR_BUCKET_NAME. Save changes.
- **Test Access:** Paste the Bucket website endpoint (found under Properties) into a browser to confirm the site is live.
- **Step 3: Implement Lifecycle Management**
- **Create Rule:** Go to the Management tab, find Lifecycle rules, and click "Create lifecycle rule".

- Define Scope: Give the rule a name (e.g., Archive-and-Expire) and choose "Apply to all objects" or use a prefix (e.g., backups/) to target specific folders.
- Configure Transitions (Cost Optimization): Check "Transition current versions of objects...".
- Click "Add transition": Move objects to Standard-IA after 30 days.
- Click "Add transition": Move objects to Glacier Instant Retrieval after 90 days.
- Configure Expiration (Cleanup): Check "Expire current versions of objects" and set the number of days after creation (e.g., 365 days) to permanently delete the objects.
- Finalize: Click "Create rule". The lifecycle rule will now automatically manage the storage class and eventual deletion of your objects based on their age.

IMPLEMENTATION SCREENSHOT



Create Lifecycle rule - S3 bucket

us-east-1.console.aws.amazon.com/s3/management/vijay-s3-buck/lifecycle/create?region=us-east-1

Search [Alt+S] United States (N. Virginia) Account ID: 2482-6826-5329 Vijay

Amazon S3 > Buckets > vijay-s3-buck > Lifecycle configuration > Create lifecycle rule

Create lifecycle rule Info

Lifecycle rule configuration

Lifecycle rule name

vijay-lifecycle-rule-1

Up to 255 characters

Choose a rule scope

☐ Limit the scope of this rule using one or more filters

☒ Apply to all objects in the bucket

⚠️ Apply to all objects in the bucket

If you want the rule to apply to specific objects, you must use a filter to identify those objects. Choose "Limit the scope of this rule using one or more filters". [Learn more](#)

☒ I acknowledge that this rule will apply to all objects in the bucket.

Lifecycle rule actions

Choose the actions you want this rule to perform.

☒ Transition current versions of objects between storage classes

This action will move current versions.

☐ Transition noncurrent versions of objects between storage classes

CloudShell

Feedback

Console Mobile App

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Create Lifecycle rule - S3 bucket

us-east-1.console.aws.amazon.com/s3/management/vijay-s3-buck/lifecycle/create?region=us-east-1

Search [Alt+S] United States (N. Virginia) Account ID: 2482-6826-5329 Vijay

Amazon S3 > Buckets > vijay-s3-buck > Lifecycle configuration > Create lifecycle rule

Day 0

• Objects uploaded

↓

Day 30

• Objects move to Standard-IA

↓

Day 60

• Objects move to Intelligent-Tiering

↓

Day 90

• Objects move to One Zone-IA

↓

Day 240

• Objects move to Glacier Flexible Retrieval (formerly Glacier)

↓

Day 365

• Objects move to Glacier Deep Archive

Day 0

No actions defined.

Cancel

Create rule

CloudShell

Feedback

Console Mobile App

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Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions

Choose storage class transitions	Days after object creation	
Standard-IA	30	<button>Remove</button>
Intelligent-Tiering	60	<button>Remove</button>
One Zone-IA	90	<button>Remove</button>
Glacier Flexible Retrieval (formerly Glacier)	240	<button>Remove</button>
Glacier Deep Archive	365	<button>Remove</button>
<button>Add transition</button>		

Lifecycle configuration

To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Default minimum object size for transitions
All storage classes 128K

Lifecycle rules (1)

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Find lifecycle rules by name

	Lifecycle rule name	Status	Scope	Current version actions	Noncurrent versions ac...	Expired object delete ...
☐	vijay-lifecycle-rule-1	Enabled	Entire bucket	Transition to Standard-IA, then I	-	-

RESULT

Thus, a static website is deployed in S3 and it is managed using S3 life cycle management.

EXP.NO: 03	Creating DNS records, configuring routing policies, implementing health checks, and monitoring DNS health using Route 53
DATE: 12-11-25	

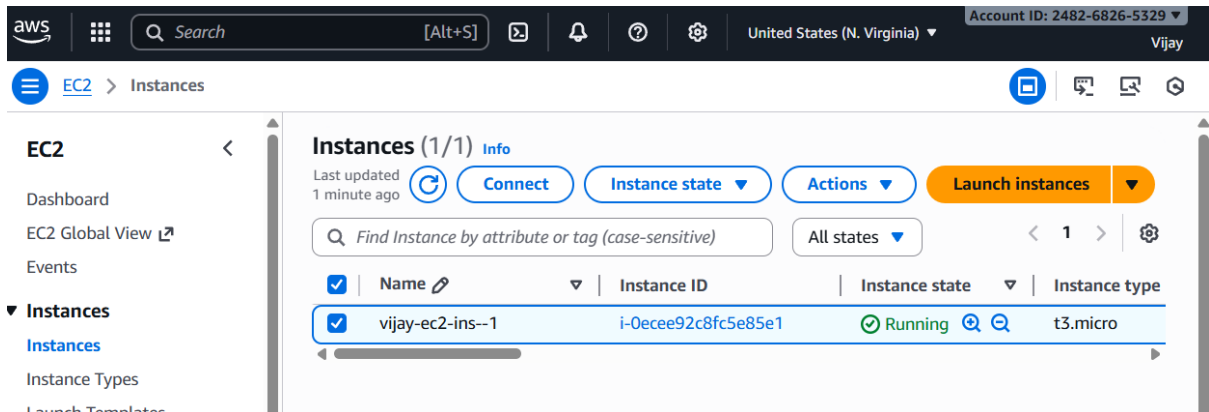
AIM

To establish a low-latency, and highly available Domain Name System (DNS) architecture for a web application using AWS Route 53, including the configuration of various routing policies, implementation of active health checks for continuous resource monitoring, and integration with CloudWatch for automated DNS health monitoring and alerting capabilities.

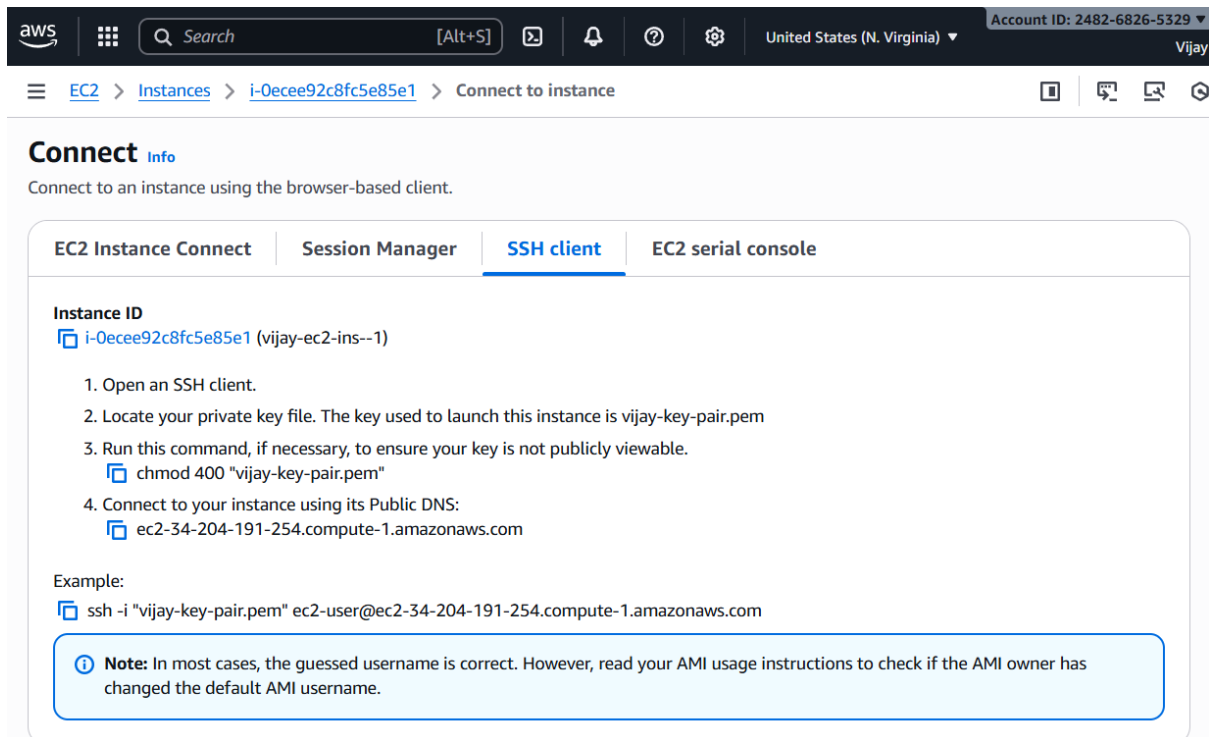
PROCEDURE

1. Create Hosted Zone: In Route 53, create a Public Hosted Zone for your domain (example.com).
2. Create Resource Records: Create an A record (or CNAME/ALIAS) pointing to your resource (e.g., EC2/Load Balancer IP). Set the Routing Policy (e.g., Simple or Failover).
3. Implement Health Check: Create a Health Check to monitor your endpoint (IP address, domain, port, and protocol, e.g., HTTP on port 80).
4. Configure Routing: For Failover or Weighted routing, edit the resource record and associate it with the new Health Check.
5. Set Monitoring/Alerts: Select the Health Check and create a CloudWatch Alarm (e.g., using the HealthCheckStatus metric) to send an SNS notification if the status becomes Unhealthy.

IMPLEMENTATION SCREENSHOT



The screenshot shows the AWS Management Console interface for the EC2 service. The top navigation bar includes the AWS logo, a search bar, and the account ID '2482-6826-5329'. The left sidebar shows the 'EC2' menu with options like 'Dashboard', 'EC2 Global View', 'Events', 'Instances', 'Instance Types', and 'Launch Templates'. The main content area is titled 'Instances (1/1)' and shows a table with one instance: 'vijay-ec2-ins--1' with ID 'i-0ecee92c8fc5e85e1', state 'Running', and type 't3.micro'. The instance is highlighted in blue.



The screenshot shows the 'Connect' page in the AWS Management Console. The page title is 'Connect' and it includes a sub-header 'Info'. The main content area is titled 'Connect to instance' and provides instructions on how to connect to the instance using an SSH client. The instructions include the following steps:

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is vijay-key-pair.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 "vijay-key-pair.pem"`
4. Connect to your instance using its Public DNS:
`ec2-34-204-191-254.compute-1.amazonaws.com`

The page also includes an 'Example' section with the following command:

```
ssh -i "vijay-key-pair.pem" ec2-user@ec2-34-204-191-254.compute-1.amazonaws.com
```

A note at the bottom states: 'Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.'


```
[root@ip-172-31-72-108 ec2-user]# yum install httpd -y
Amazon Linux 2023 Kernel Livepatch repository                               278 kB/s | 29 kB    00:00
Dependencies resolved.
=====
Package                               Architecture      Version                               Repository          Size
=====
Installing:
httpd                                  x86_64            2.4.65-1.amzn2023.0.2               amazonlinux          47 k
Installing dependencies:
apr                                    x86_64            1.7.5-1.amzn2023.0.4               amazonlinux          129 k
apr-util                              x86_64            1.6.3-1.amzn2023.0.2               amazonlinux          97 k
apr-util-ldap                         x86_64            1.6.3-1.amzn2023.0.2               amazonlinux          13 k
generic-logos-httpd                  noarch            18.0.0-12.amzn2023.0.3             amazonlinux          19 k
httpd-core                            x86_64            2.4.65-1.amzn2023.0.2               amazonlinux          1.4 M
httpd-filesystem                     noarch            2.4.65-1.amzn2023.0.2               amazonlinux          13 k
httpd-tools                           x86_64            2.4.65-1.amzn2023.0.2               amazonlinux          81 k
libbrotli                             x86_64            1.0.9-4.amzn2023.0.2               amazonlinux          315 k
mailcap                               noarch            2.1.49-3.amzn2023.0.3             amazonlinux          33 k
Installing weak dependencies:
apr-util-openssl                     x86_64            1.6.3-1.amzn2023.0.2               amazonlinux          15 k
mod_http2                             x86_64            2.0.27-1.amzn2023.0.3             amazonlinux          166 k
mod_lua                               x86_64            2.4.65-1.amzn2023.0.2               amazonlinux          60 k

Transaction Summary
=====
Install 13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm                2.7 MB/s | 97 kB    00:00
(2/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm           346 kB/s | 13 kB    00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                     2.9 MB/s | 129 kB    00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm        734 kB/s | 15 kB    00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm   822 kB/s | 19 kB    00:00
(6/13): httpd-2.4.65-1.amzn2023.0.2.x86_64.rpm                 2.1 MB/s | 47 kB    00:00
(7/13): httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch.rpm       604 kB/s | 13 kB    00:00
(8/13): httpd-tools-2.4.65-1.amzn2023.0.2.x86_64.rpm           3.5 MB/s | 81 kB    00:00
(9/13): httpd-core-2.4.65-1.amzn2023.0.2.x86_64.rpm            33 MB/s | 1.4 MB    00:00
(10/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm               1.5 MB/s | 33 kB    00:00
(11/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm             8.5 MB/s | 315 kB   00:00
```

```
Installing : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64           6/13
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64             7/13
Running scriptlet: httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch 8/13
Installing : httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch     8/13
Installing : httpd-core-2.4.65-1.amzn2023.0.2.x86_64          9/13
Installing : mod_http2-2.0.27-1.amzn2023.0.3.x86_64           10/13
Installing : mod_lua-2.4.65-1.amzn2023.0.2.x86_64             11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.65-1.amzn2023.0.2.x86_64               13/13
Running scriptlet: httpd-2.4.65-1.amzn2023.0.2.x86_64         13/13
Verifying   : apr-1.7.5-1.amzn2023.0.4.x86_64                 1/13
Verifying   : apr-util-1.6.3-1.amzn2023.0.2.x86_64            2/13
Verifying   : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64       3/13
Verifying   : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64    4/13
Verifying   : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying   : httpd-2.4.65-1.amzn2023.0.2.x86_64              6/13
Verifying   : httpd-core-2.4.65-1.amzn2023.0.2.x86_64        7/13
Verifying   : httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch    8/13
Verifying   : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64       9/13
Verifying   : libbrotli-1.0.9-4.amzn2023.0.2.x86_64          10/13
Verifying   : mailcap-2.1.49-3.amzn2023.0.3.noarch            11/13
Verifying   : mod_http2-2.0.27-1.amzn2023.0.3.x86_64         12/13
Verifying   : mod_lua-2.4.65-1.amzn2023.0.2.x86_64           13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64                                apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64                    apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch             httpd-2.4.65-1.amzn2023.0.2.x86_64
httpd-core-2.4.65-1.amzn2023.0.2.x86_64                       httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch
httpd-tools-2.4.65-1.amzn2023.0.2.x86_64                     libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch                           mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.65-1.amzn2023.0.2.x86_64                          mod_lua-2.4.65-1.amzn2023.0.2.x86_64

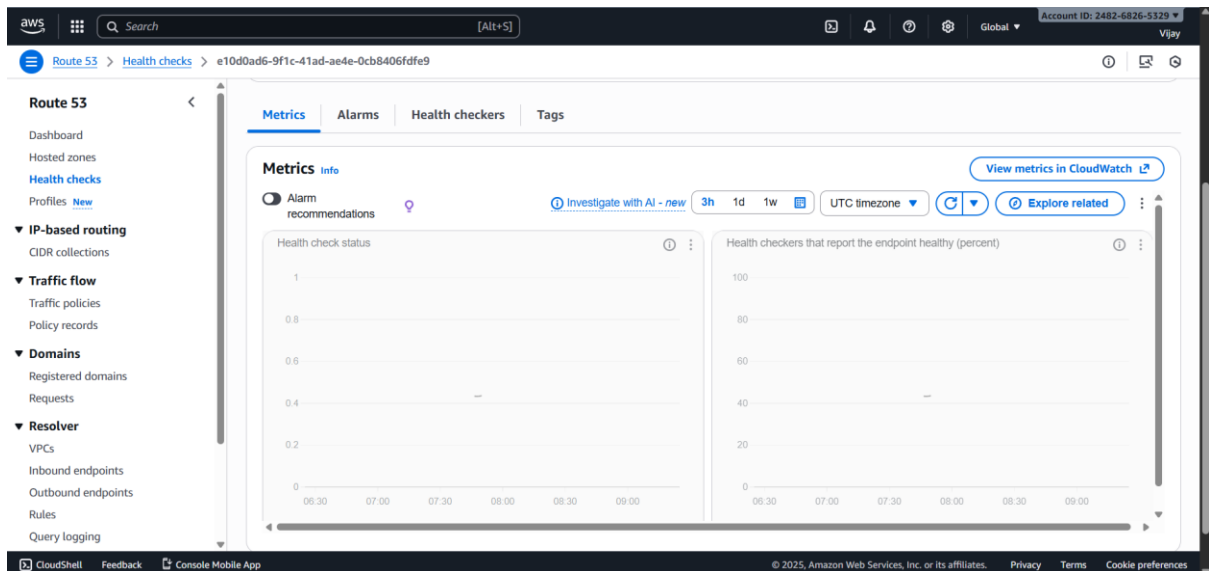
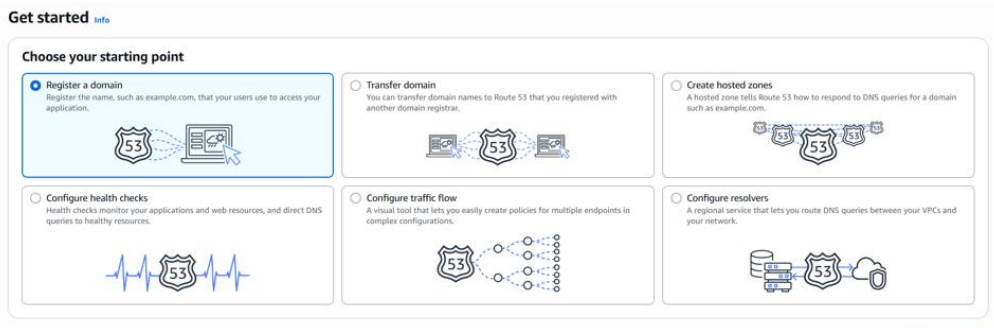
Complete!
```

```
[root@ip-172-31-72-108 ec2-user]# sudo systemctl start httpd
[root@ip-172-31-72-108 ec2-user]# sudo systemctl enable httpd
```

```
[root@ip-172-31-72-108 ec2-user]# sudo systemctl status httpd
```

```
[root@ip-172-31-72-108 ec2-user]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Fri 2025-11-14 06:11:07 UTC; 21min ago
     Docs: man:httpd.service(8)
  Main PID: 25542 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1053)
    Memory: 13.3M
       CPU: 1.166s
    CGroup: /system.slice/httpd.service
            └─25542 /usr/sbin/httpd -DFOREGROUND
              └─25543 /usr/sbin/httpd -DFOREGROUND
                └─25544 /usr/sbin/httpd -DFOREGROUND
                  └─25545 /usr/sbin/httpd -DFOREGROUND
                    └─25546 /usr/sbin/httpd -DFOREGROUND

Nov 14 06:11:07 ip-172-31-72-108.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Nov 14 06:11:07 ip-172-31-72-108.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 14 06:11:07 ip-172-31-72-108.ec2.internal httpd[25542]: Server configured, listening on: port 80
[root@ip-172-31-72-108 ec2-user]#
```



RESULT

Thus, successfully deployed and configured is a foundational web infrastructure: a dynamic EC2/Apache server is operational alongside a scalable, cost-optimized S3 static hosting platform, all centrally managed by AWS Route 53 which provides the domain name, intelligent routing policies, and proactive health checks to ensure high availability and automated failover capabilities while S3 Lifecycle Management optimizes storage costs.

EXP.NO: 04	Create and configure Amazon Elastic File System (EFS) to create scalable and shared file storage for AWS resources and EFS file system, mount it to EC2 instances, and configure access permissions
DATE: 13-11-25	

AIM

To successfully create, configure, and securely mount an Amazon Elastic File System (EFS) file system onto one or more Amazon EC2 (Linux) instances within the same Virtual Private Cloud (VPC), thereby establishing a highly available, scalable, and shared network file storage solution for distributed applications. The aim also includes setting up access control using Security Groups and, optionally, EFS Access Points and IAM policies.

PROCEDURE

1. Create EFS File System: Navigate to the EFS console, select "Create file system," choose your target VPC, and configure performance/storage settings.
2. Define Mount Targets: When creating the EFS, select the appropriate Availability Zones (AZs) to automatically create Mount Targets and attach a new or existing Security Group (e.g., EFS-SG) to these targets.
3. Configure EFS Security Group (EFS-SG): Edit the inbound rules of the EFS-SG to explicitly allow NFS traffic (Port 2049), specifying the Security Group of your EC2 instance(s) as the source.
4. Verify EC2 Security Group (EC2-SG): Ensure the EC2-SG has an outbound rule that permits all traffic (or specifically Port 2049) destined for the EFS-SG to allow the connection initiation.
5. Connect and Install Utility: SSH into your EC2 instance and install the recommended Amazon EFS client utility (amazon-efs-utils) or the generic nfs-common package if using a non-Amazon Linux distribution.

6. Create Mount Point: Create a local directory on the EC2 instance to serve as the mount point (e.g., /mnt/efs_share).
7. Mount the File System: Use the EFS Mount Helper command (found in the EFS console's "Attach" instructions) to execute the mount command: `sudo mount -t efs -o tls fs-xxxxxxx:/ /mnt/efs_share`.

IMPLEMENTATION SCREENSHOT

The top screenshot displays the AWS Management Console's EC2 Instances page. A table lists instances, with the following data visible:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
23bai10917...	i-092f429fc8cc79424	Running	t3.micro	Initializing	View alarms +	us-east-1f	ec2-44-210-134-91.co...	44.210

The bottom screenshot shows the Amazon EFS File systems page. A table lists file systems, with the following data visible:

Name	File system ID	Encrypte d	Total size	Size in Standard	Size in IA	Size in Archive	Provisioned Throughput (MiB/s)	File system state	Creation time
23bai10917-elastic-file-system	fs-09eaabfd43cd3ab6e	Encrypte d	0 Bytes	0 Bytes	0 Bytes	0 Bytes	-	Creating	Wed, 12 Nov 2025 15:51:48 GMT

23bai10917-elastic-file-system (fs-09eaabfd43cd3ab6e)

General

Amazon resource name (ARN)
arn:aws:elasticfilesystem:us-east-1:248268265329:file-system/fs-09eaabfd43cd3ab6e

Automatic backups
Enabled

Encrypted
3b6ce6f1-7392-4310-9fec-f1cef1cd927 (aws/elasticfilesystem)

File system state
Available

DNS name
fs-09eaabfd43cd3ab6e.efs.us-east-1.amazonaws.com

Replication overwrite protection
Enabled

Metered size
Total size
6.00 KiB

Metered size

Total size
6.00 KiB

Size in Standard
6.00 KiB (100%)

Size in IA
0 Bytes (0%)

Size in Archive
0 Bytes (0%)

6.00 KiB

Size in Standard
6.00 KiB

Size in Standard Size in IA Size in Archive

Attach

Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

☒ Mount via DNS ☐ Mount via IP

Using the EFS mount helper:

```
sudo mount -t efs -o tls fs-09eaabfd43cd3ab6e:/ efs
```

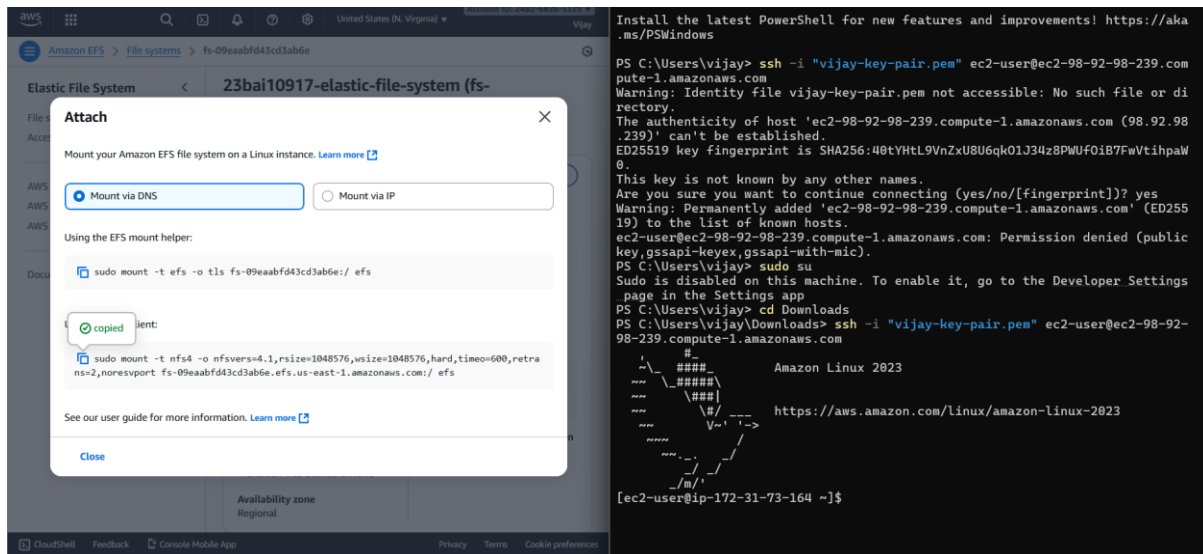
Using the NFS client:

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsizex=1048576,hard,timeo=600,retrns=2,noresvport fs-09eaabfd43cd3ab6e.efs.us-east-1.amazonaws.com:/ efs
```

See our user guide for more information. [Learn more](#)

Close

Metered size
Total size
6.00 KiB



RESULT

The EFS file system will be successfully mounted to the EC2 instance, verified by the presence of the EFS network volume when running the `df -hT` command. This result confirms that the storage is decoupled from the compute instance, allowing files written to the mounted directory to be immediately accessible, scalable, and persistent across all other EC2 instances securely mounted to the same EFS.

EXP.NO: 05	Create and configure CloudFront distributions to accelerate the delivery of their web content, improve website performance, and enhance user experience
DATE: 13-11-25	

AIM

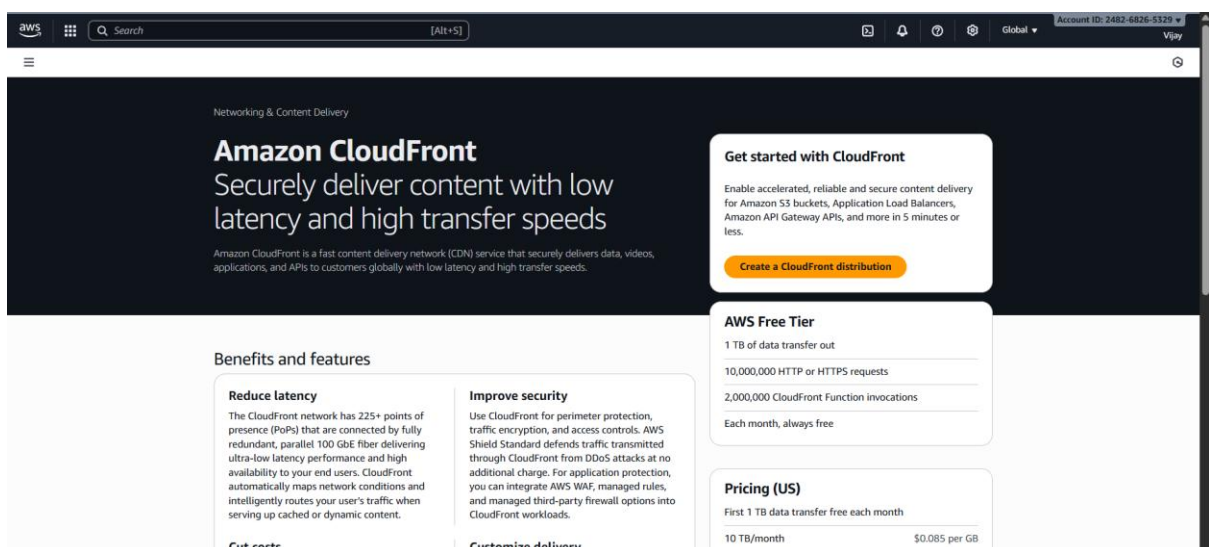
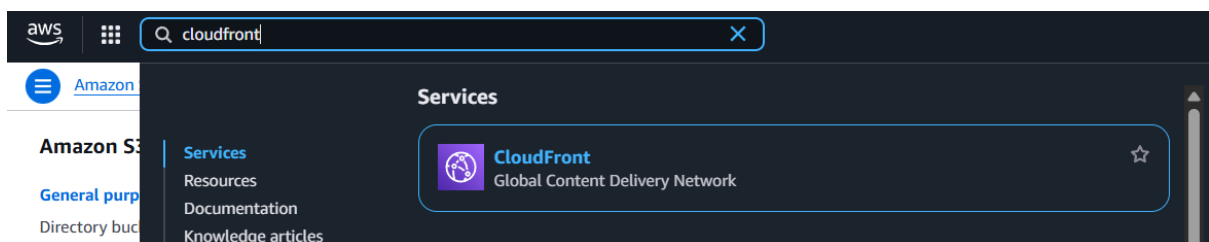
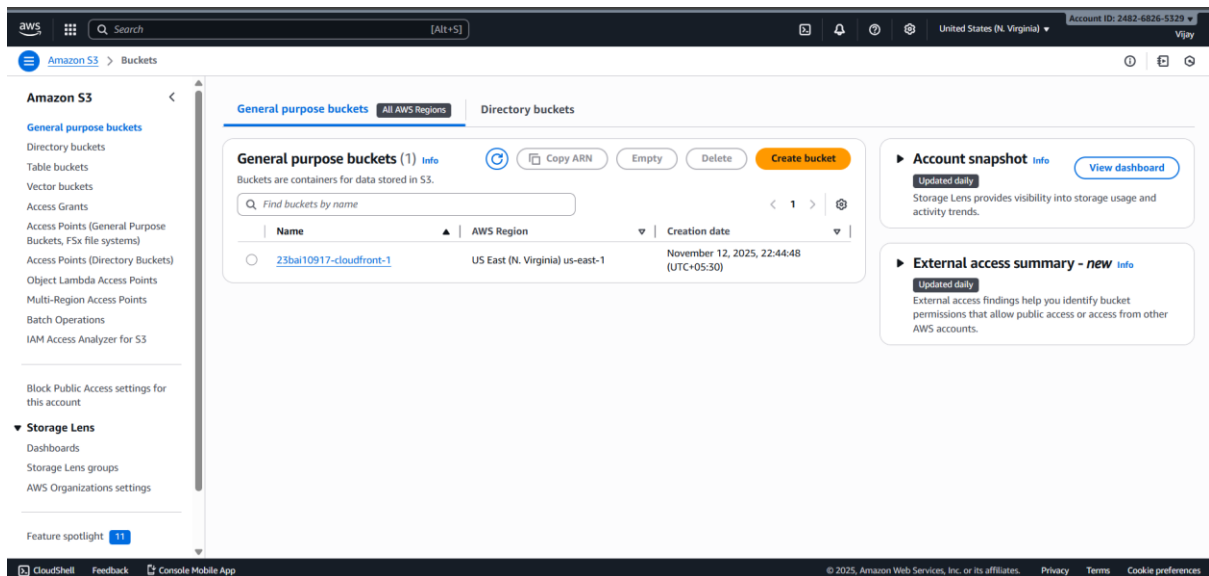
To Create and configure CloudFront distributions to accelerate the delivery of their web content, improve website performance, and enhance user experience.

PROCEDURE

1. Select Distribution Type: Navigate to the CloudFront console and choose "Create Distribution," selecting the appropriate delivery method (usually Web for website content).
2. Define Origin: Specify the Origin Domain Name (e.g., your S3 bucket endpoint or Load Balancer DNS for EC2), set a unique Origin ID, and configure an Origin Access Control (OAC) to restrict direct public access to the Origin.
3. Configure Cache Behavior: Define the path patterns (e.g., Default (*)) and set the Viewer Protocol Policy (e.g., Redirect HTTP to HTTPS) and Allowed HTTP Methods (e.g., GET, HEAD).
4. Set Caching Policy: Choose or create a Cache Policy to control how long content stays in the edge locations (Time-To-Live or TTL) and which request headers, cookies, or query strings are forwarded to the Origin.
5. Distribution Settings: Specify the Price Class (geographic regions to serve), the Alternate Domain Names (CNAMEs) for your custom domain, and the Custom SSL Certificate for secure connections.
6. Create and Wait: Review all settings and click "Create Distribution." Wait for the distribution's status to change from InProgress to Deployed (this typically takes 15–30 minutes).
7. Update DNS: Once deployed, update your domain's DNS records by creating a CNAME record that points your custom domain name (e.g., www.example.com) to the new CloudFront Distribution Domain Name.

IMPLEMENTATION SCREENSHOT

Creating S3 bucket.



26

Successfully created new distribution.

23bai10917-cloudfront-distribution Standard View metrics

Details

Distribution domain name duo4nz3fjpngx.cloudfront.net	ARN arn:aws:cloudfront::248268265329:distribution/E1OH9NKFTL28IE	Last modified Deploying
--	---	----------------------------

General | Security | Origins | Behaviors | Error pages | Invalidations | Tags | Logging

Settings Edit

Name 23bai10917-cloudfront-distribution	Alternate domain names Add domain	Standard logging Off
Description 		Cookie logging Off
Price class Use all edge locations (best performance)		Default root object
Supported HTTP versions HTTP/2, HTTP/1.1, HTTP/1.0		

CloudFront > Distributions > EBP8YJN3KMYV > Edit origin

CloudFront

Distributions
Policies
Functions
Static IPs
VPC origins
What's new

Bucket can restrict access to only CloudFront.

☐ Legacy access identities
Use a CloudFront origin access identity (OAI) to access the S3 bucket.

Origin access control
Select an existing origin access control (recommended) or create a new control.

23bai10917-cloudfront-1.s3.us-east-1.amazonaws.com Create new OAC

You must allow access to CloudFront using this policy statement. Learn more about giving CloudFront permission to access the S3 bucket. Copy policy

Amazon S3 > Buckets > 23bai10917-cloudfront-1 > Edit bucket policy

Amazon S3

General purpose buckets
Directory buckets
Table buckets
Vector buckets
Access Grants
Access Points (General Purpose Buckets, FSx file systems)
Access Points (Directory Buckets)
Object Lambda Access Points
Multi-Region Access Points
Batch Operations
IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens
Dashboards
Storage Lens groups
AI/ML Recommendation settings

Bucket ARN
arn:aws:s3::23bai10917-cloudfront-1

Policy

```

1  {
2    "Version": "2008-10-17",
3    "Id": "PolicyForCloudFrontPrivateContent",
4    "Statement": [
5      {
6        "Sid": "AllowCloudFrontServicePrincipal",
7        "Effect": "Allow",
8        "Principal": {
9          "Service": "cloudfront.amazonaws.com"
10       },
11       "Action": "s3:GetObject",
12       "Resource": "arn:aws:s3:::23bai10917-cloudfront-1/*",
13       "Condition": {
14         "StringEquals": {
15           "AWS:SourceArn": "arn:aws:cloudfront::248268265329:distribution/EBP8YJN3KMYV"
16         }
17       }
18     ]
19   }
20 }

```

Edit statement

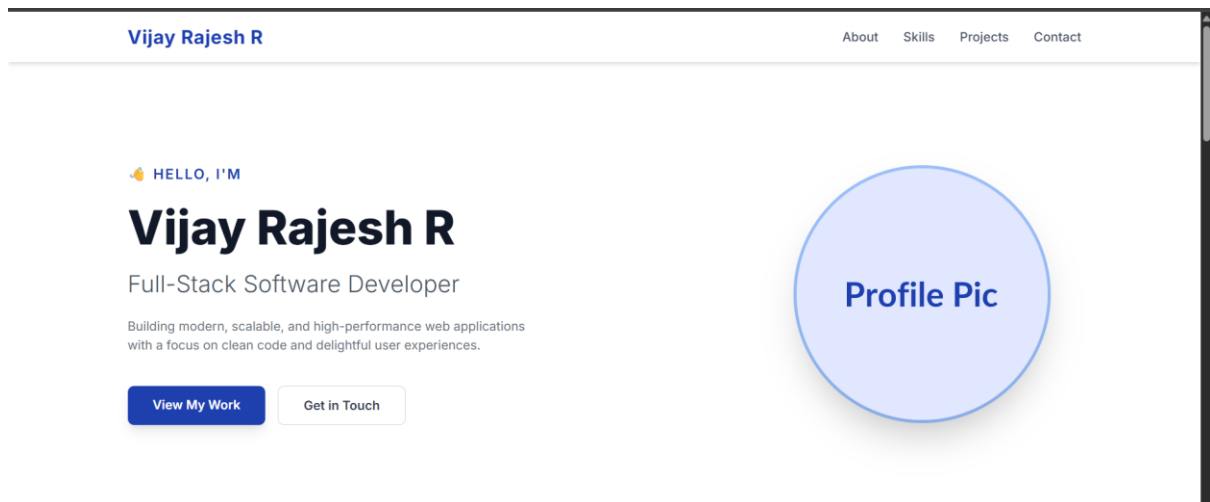
Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

23bai10917-cloudfront-1

Successfully edited bucket policy.



RESULT

Upon updating the DNS and successful propagation, website visitors accessing the custom domain will be automatically routed to the nearest CloudFront Edge Location, resulting in reduced latency, faster loading times, and a highly available content delivery experience. The distribution will effectively cache content, minimizing direct requests to the Origin and improving overall website performance and security.

EXP.NO: 06	Create AWS Certificate Manager (ACM) for managing SSL/TLS certificates for secure communication in AWS
DATE: 14-11-25	

AIM

To **request a public SSL/TLS certificate** from AWS Certificate Manager (ACM), **validate domain ownership** using DNS records, and **integrate** the issued certificate with an **Application Load Balancer (ALB)** to enable secure **HTTPS communication** for a hosted application.

PROCEDURE

1. Request an ACM Certificate: Navigate to the ACM console and choose Request a public certificate. Enter the desired fully qualified domain name (FQDN) (e.g., www.example-lab.com) and a wildcard domain (e.g., *.example-lab.com) for multi-subdomain support. Select the DNS validation method and click Request.
2. Validate Domain Ownership (DNS): Once the certificate request is in Pending validation status, retrieve the CNAME record details provided by ACM. If the domain's DNS is hosted in Amazon Route 53, use the "Create records in Route 53" button to automatically add the required CNAME record to the public hosted zone. Wait for the certificate status to change to Issued.
3. Provision Target EC2 Instance: Launch a simple EC2 instance running a web server (like Apache or NGINX) to serve plain HTTP content on port 80. Create a Target Group in EC2 to register this instance as a target.
4. Create Application Load Balancer (ALB): Create an Application Load Balancer with listeners configured for both HTTP (port 80) and HTTPS (port 443). Configure the HTTP listener to redirect traffic to HTTPS.
5. Integrate ACM Certificate with ALB: For the HTTPS (port 443) listener on the ALB, under Default SSL/TLS certificate, select Choose a certificate from ACM and choose the Issued certificate from Step 2. Select a secure Security policy (e.g., ELBSecurityPolicy-2016-08).
6. Configure DNS Record for ALB: In Route 53, create an Alias record for the FQDN (www.example-lab.com) that points directly to the newly created ALB DNS name. This directs all public traffic for the domain to the Load Balancer.
7. Final Validation: Access the domain in a web browser using HTTPS (e.g., https://www.example-lab.com). Verify that the browser displays a secure connection (lock icon) and confirm the certificate details show the ACM-issued certificate. Attempting to access the

HTTP URL should automatically redirect to the secure HTTPS connection.

IMPLEMENTATION SCREENSHOT

The top screenshot displays the AWS Route 53 console for the hosted zone 'example-23bai10917.com'. The 'Records' tab is active, showing a table of DNS records. The table has columns for Record, Type, Routing, Differ, Alias, Value/Route traffic to, TTL, Health, and Evaluation. Three records are listed: an A record for 'example-...' pointing to '192.0.2.235', an NS record for 'example-...' pointing to 'ns-1896.awsdns-45.co.uk', and a SOA record for 'example-...' pointing to 'ns-1896.awsdns-45.co.uk'. The bottom screenshot shows the AWS Certificate Manager (ACM) console. The 'New ACM managed certificate' section is highlighted, showing options to 'Request a certificate', 'Import a certificate', and 'Create a private CA'. The 'Request a certificate' button is highlighted in orange.

Record	Type	Routing	Differ	Alias	Value/Route traffic to	TTL	Health	Evaluation
example-...	A	Simple	-	No	192.0.2.235	300	-	-
example-...	NS	Simple	-	No	ns-1896.awsdns-45.co.uk, ns-1069.awsdns-05.org, ns-166.awsdns-20.com, ns-706.awsdns-24.net	172800	-	-
example-...	SOA	Simple	-	No	ns-1896.awsdns-45.co.uk, a...	900	-	-

Request certificate | Certificate | Bills | Billing and Cost Manager

us-east-1.console.aws.amazon.com/acm/home?region=us-east-1#/certificates/request

aws Search [Alt+S] United States (N. Virginia) Account ID: 2482-6826-5329 Vijay

AWS Certificate Manager > Certificates > Request certificate

Request certificate

Certificate type Info

ACM certificates can be used to establish secure communications access across the internet or within an internal network. Choose the type of certificate for ACM to provide.

☒ Request a public certificate

Request a public SSL/TLS certificate from Amazon. By default, public certificates are trusted by browsers and operating systems.

☐ Request a private certificate

No private CAs available for issuance.

Requesting a private certificate requires the creation of a private certificate authority (CA). To create a private CA, visit [AWS Private Certificate Authority](#)

Cancel Next

AWS Certificate Manager > Certificates > Request certificate > Request public certificate

Request public certificate

Domain names

Provide one or more domain names for your certificate.

Fully qualified domain name Info

example-23bai10917.com

Add another name to this certificate

You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.

Allow export Info

☒ Disable export

Use this certificate only with integrated AWS services. The private key for this certificate will be disallowed for exporting from AWS.

☐ Enable export

Export this certificate and private key for use with any TLS workflow. ACM will charge your account based on the requested domains when the certificate is issued for the first time and for each renewal.

Validation method Info

Select a method for validating domain ownership.

AWS Certificate Manager > Certificates > Request certificate > Request public certificate

Request public certificate

Key algorithm Info

Select an encryption algorithm. Some algorithms may not be supported by all AWS services.

☒ RSA 2048

RSA is the most widely used key type.

☐ ECDSA P 256

Equivalent in cryptographic strength to RSA 3072.

☐ ECDSA P 384

Equivalent in cryptographic strength to RSA 7680.

Tags Info

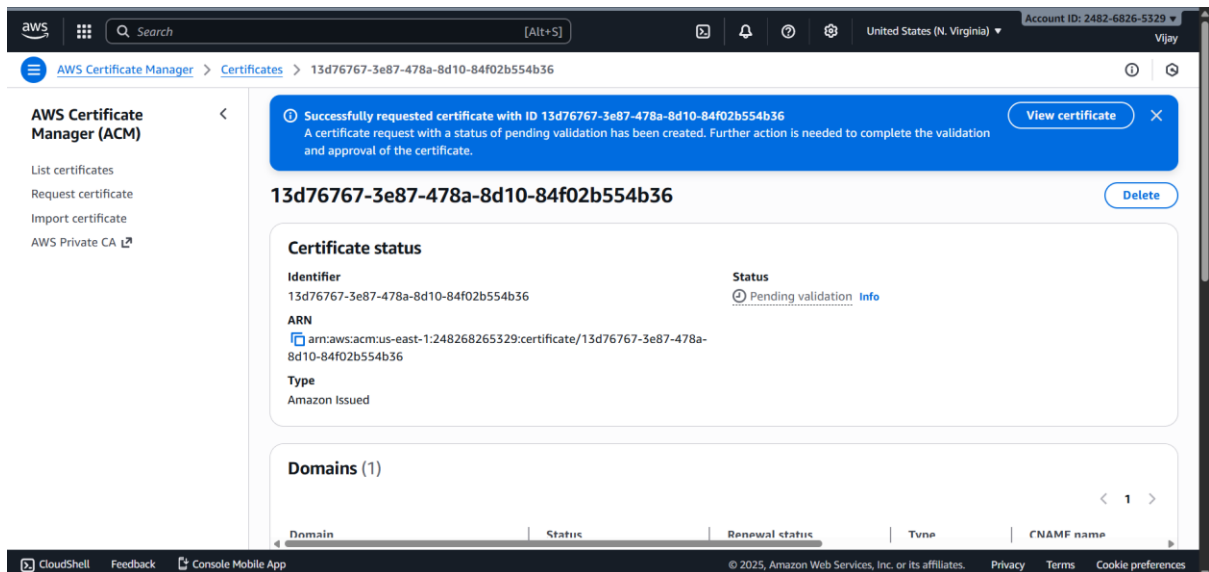
No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel Previous Request

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RESULT

The lab successfully demonstrated the automated provisioning of a publicly trusted SSL/TLS certificate via AWS Certificate Manager and its immediate integration with an Application Load Balancer. The certificate achieved the Issued status following DNS validation, and when integrated with the ALB, it enforced secure HTTPS termination for the web application, proving ACM's capability to manage secure communication with minimal manual intervention and ensuring managed certificate renewal.

EXP.NO: 07	Create and manage customer master keys (CMKs), encrypt data using KMS, and integrate KMS with AWS services for encryption
DATE: 14-11-25	

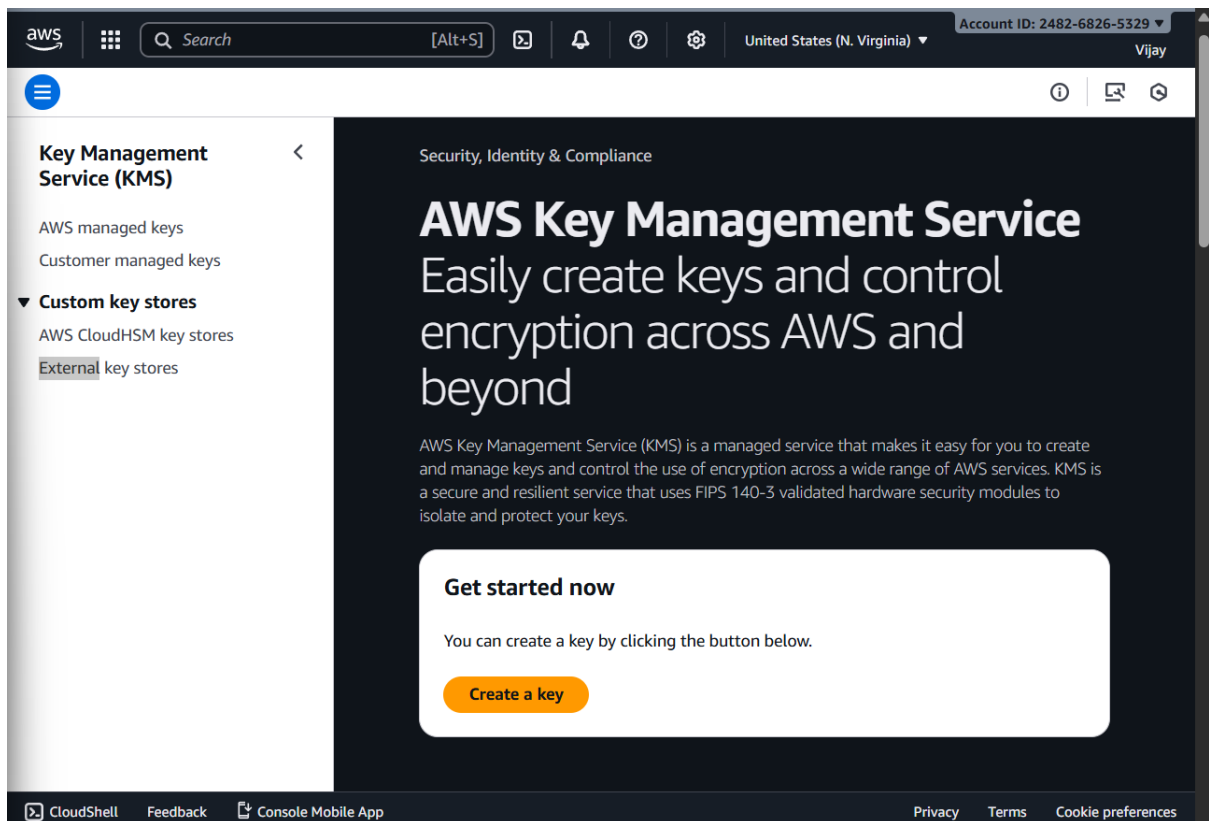
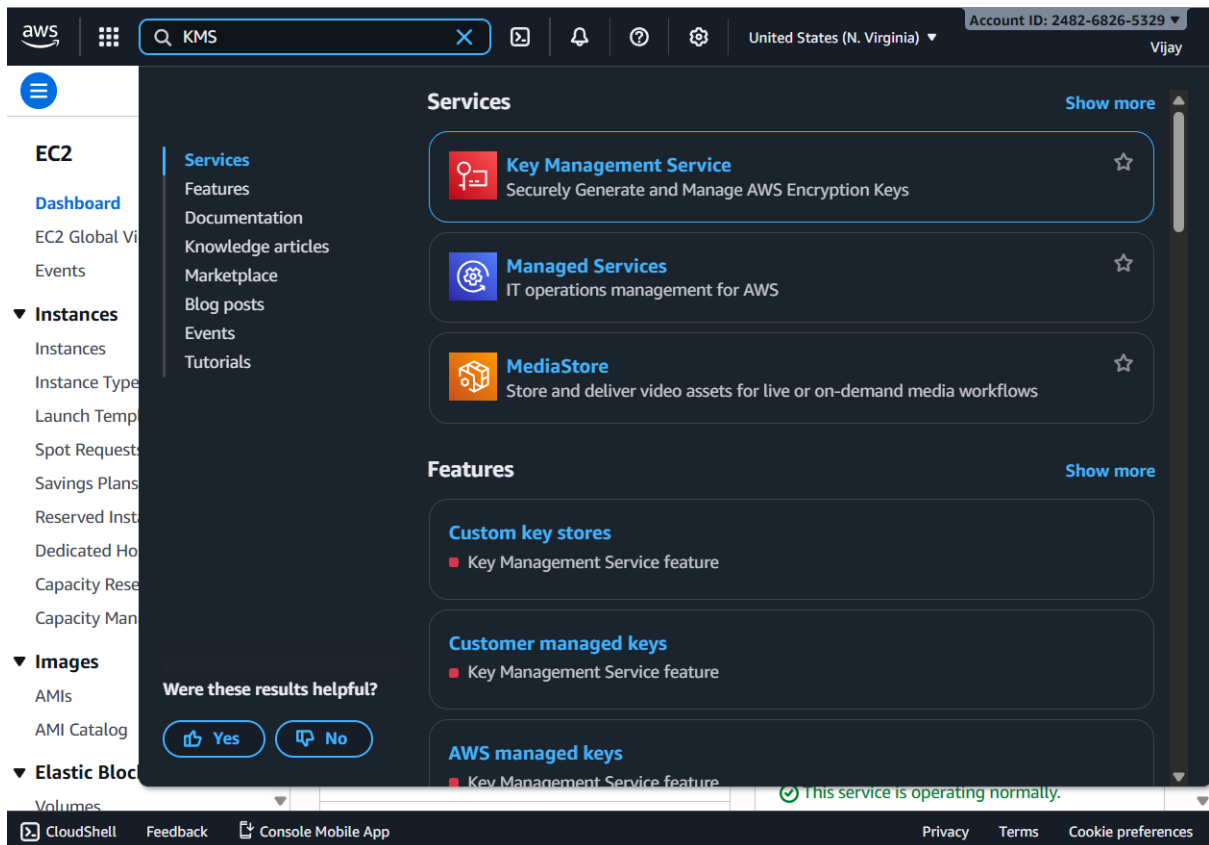
AIM

To create and manage a Customer Managed Key (CMK) in AWS KMS, use this CMK to perform envelope encryption on data, and integrate the CMK with a primary AWS service (e.g., Amazon S3) to enforce server-side encryption.

PROCEDURE

1. Create Customer Managed Key (CMK): Navigate to AWS KMS, select Customer managed keys, and choose Create key. Select Symmetric key type. Define the key administrators and key usage permissions (users/roles allowed to encrypt/decrypt data) via the key policy. Give the key a unique alias (e.g., Lab-S3-CMK) and description.
2. Create Encrypted S3 Bucket: Navigate to Amazon S3 and create a new bucket (e.g., kms-encrypted-bucket-1234). In the Default encryption settings, select Server-Side Encryption with AWS Key Management Service key (SSE-KMS). Choose the CMK created in Step 1 (Lab-S3-CMK) from the dropdown list.
3. Validate Default Encryption: Upload a new text file or image to the S3 bucket created in Step 2. Immediately check the object's Properties under the Server-side encryption section to confirm that it was automatically encrypted using SSE-KMS and the specific CMK ARN.
4. Simulate Encryption Failure (Key Disable): Navigate back to the AWS KMS console, select the CMK (Lab-S3-CMK), and choose Disable from the Key actions menu. This renders the key unusable for cryptographic operations.
5. Attempt Data Access with Disabled Key: Attempt to download the encrypted object from the S3 bucket. The download should fail with an "Access Denied" or decryption error, confirming that access to the encrypted data is directly tied to the availability and status of the CMK.
6. Re-Enable and Verify Access: Navigate to AWS KMS, select the CMK (Lab-S3-CMK), and choose Enable from the Key actions menu. Wait a moment for the state to update.
7. Final Decryption Validation: Attempt to download the encrypted object from the S3 bucket again. The download should now succeed, and the file should be fully readable, confirming successful key management and integration.

IMPLEMENTATION SCREENSHOT



aws Search [Alt+S] United States (N. Virginia) Account ID: 2482-6826-5329 Vijay

KMS > Customer managed keys

Key Management Service (KMS)

AWS managed keys

[Customer managed keys](#)

▼ **Custom key stores**

AWS CloudHSM key stores

External key stores

Customer managed keys (0) Key actions Create key

	Aliases	Key ID	Status	Key type	Key spec	Key us
Empty Resources						
No resources to display						

aws Search [Alt+S] United States (N. Virginia) Account ID: 2482-6826-5329 Vijay

KMS > [Customer managed keys](#) > Create key

Step 1

Configure key

Step 2

Add labels

Step 3 - optional

Define key administrative permissions

Step 4 - optional

Define key usage permissions

Step 5 - optional

Edit key policy

Step 6

Review

Configure key

Key type [Help me choose](#)

☒ **Symmetric**

A single key used for encrypting and decrypting data or generating and verifying HMAC codes

☐ **Asymmetric**

A public and private key pair used for encrypting and decrypting data, signing and verifying messages, or deriving shared secrets

Key usage [Help me choose](#)

☒ **Encrypt and decrypt**

Use the key only to encrypt and decrypt data.

☐ **Generate and verify MAC**

Use the key only to generate and verify hash-based message authentication codes (HMAC).

► **Advanced options**

Cancel Next

Search

[Alt+S]

United States (N. Virginia)

Account ID: 2482-6826-5329

Vijay

KMS

Customer managed keys

Create key

Step 1

Configure key

Step 2

Add labels

Step 3 - optional

Define key administrative permissions

Step 4 - optional

Define key usage permissions

Step 5 - optional

Edit key policy

Step 6

Review

Add labels

Alias

You can change the alias at any time. [Learn more](#)

Alias

23bai10917--kms--key--1

Description - optional

You can change the description at any time.

Description

practice lab session

KMS

Customer managed keys

Create key

Key policy

Edit

Review the key policy for this key. Select **Edit** to make changes.

```

1 {
2   "Id": "key-consolepolicy-3",
3   "Version": "2012-10-17",
4   "Statement": [
5     {
6       "Sid": "Enable IAM User Permissions",
7       "Effect": "Allow",
8       "Principal": {
9         "AWS": "arn:aws:iam::248268265329:root"
10      },
11       "Action": "kms:*",
12       "Resource": "*"
13     }
14   ]
15 }

```

Cancel

Previous

Finish

CloudShell

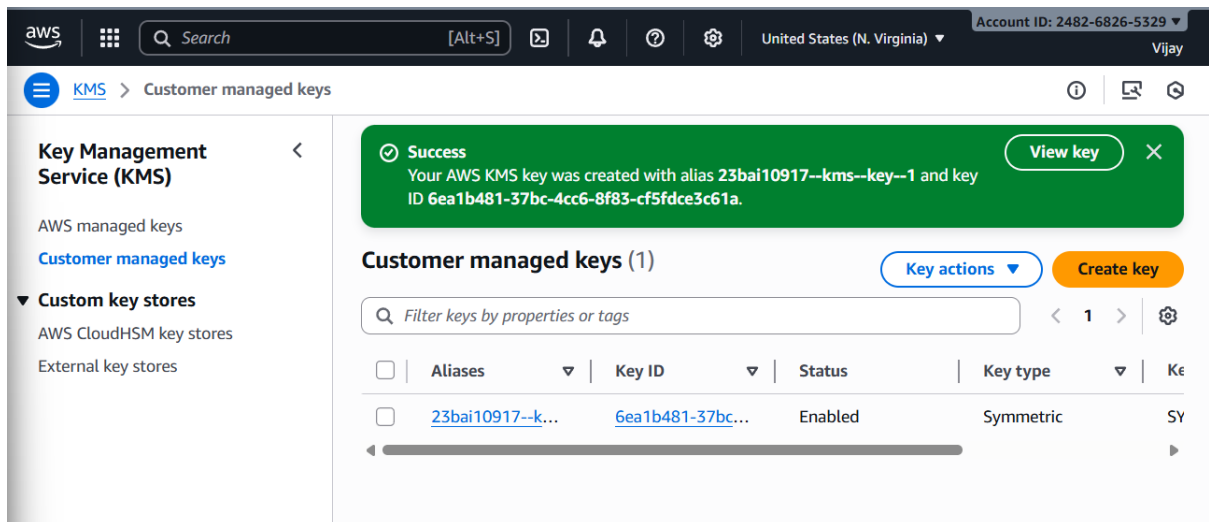
Feedback

Console Mobile App

Privacy

Terms

Cookie preferences

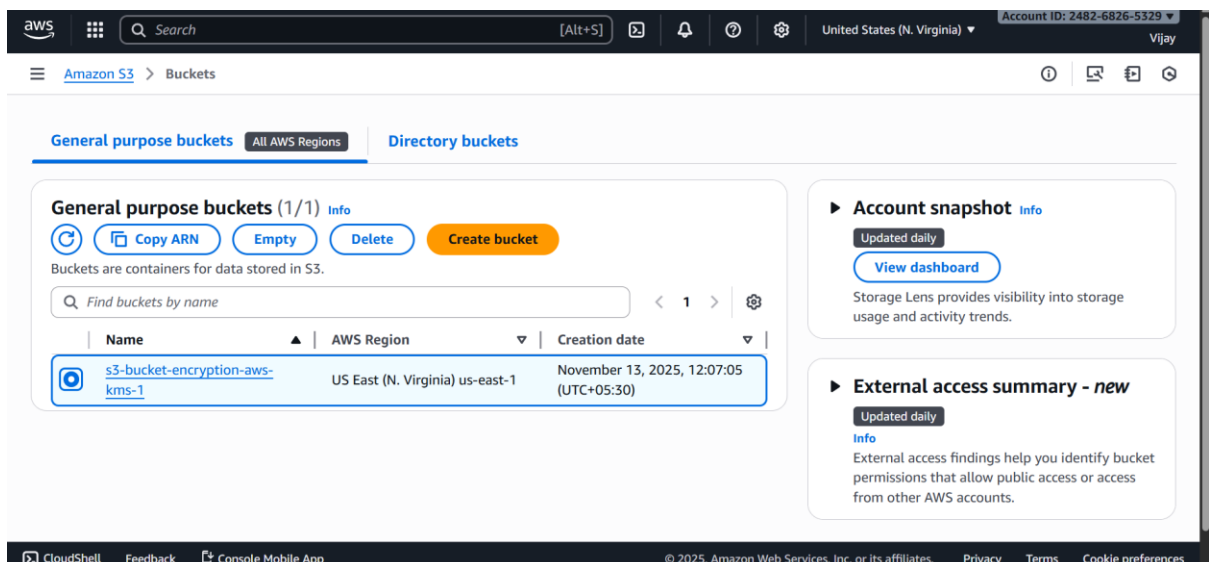


....

→ The key is created successfully... Now, integrating KMS with an AWS Service for encryption.

→ Configure an S3 bucket to use your newly created KMS key for Server-Side Encryption (SSE-KMS).

....



aws

110%United States (N. Virginia)

Account ID: 2482-6826-5329

Vijay

Amazon S3> Buckets> s3-bucket-encryption-aws-kms-1

s3-bucket-encryption-aws-kms-1

Info

ObjectsMetadataPropertiesPermissionsMetricsManagementAccess Points

Bucket overview

AWS Region

US East (N. Virginia) us-east-1

Amazon Resource Name (ARN)

arn:aws:s3:::s3-bucket-encryption-a

ws-kms-1

Creation date

November 13, 2025, 12:07:05 (UTC+05:30)

Bucket Versioning

Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

Disabled

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

CloudShellFeedbackConsole Mobile AppPrivacyTermsCookie preferences

Default encryption

[Edit](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

Server-side encryption with Amazon S3 managed keys (SSE-S3)

Bucket Key

When KMS encryption is used to encrypt new objects in this bucket, the bucket key reduces encryption costs by lowering calls to AWS KMS. [Learn more](#)

Enabled

aws

Search

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United States (N. Virginia)

Account ID: 2482-6826-5329

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Amazon S3

Buckets

s3-bucket-encryption-aws-kms-1

Edit default encryption

Edit default encryption

Info

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Info

Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the [Storage](#) tab of the [Amazon S3 pricing page](#).

☐ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☒ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

AWS KMS key

Info

☒ Choose from your AWS KMS keys

☐ Enter AWS KMS key ARN

Available AWS KMS keys

Choose AWS KMS key

arn:aws:kms:us-east-1:248268265329:key/6ea1b481-37bc-4cc6-8f83-cf5fdce3c61a

23bai10917--kms--key--1

Create a KMS key

Available AWS KMS keys

arn:aws:kms:us-east-1:248268265329:key/6ea1b481-37bc...

arn:aws:kms:us-east-1:248268265329:key/6ea1b481-37bc-4cc6-8f83-cf5fdce3c61a

23bai10917--kms--key--1

aws

U

Account ID: 2482-6826-5329

Vijay

Amazon S3

Buckets

s3-bucket-encryption-aws-kms-1

Successfully edited default encryption. Objects uploaded, modified, or copied into this bucket will inherit this encryption configuration unless otherwise specified.

s3-bucket-encryption-aws-kms-1

Info

RESULT

The successful execution of the lab confirmed that the Customer Managed Key (CMK) was created and its usage permissions defined via the key policy. Integration with S3 was validated by observing automatic SSE-KMS encryption on all uploaded objects. Crucially, disabling the CMK instantly revoked access to the encrypted S3 objects, and re-enabling the CMK immediately restored access, demonstrating the centralized control and granular security provided by AWS KMS over data encryption.

EXP.NO: 08	Set up CloudWatch monitoring for AWS resources, create custom metrics and alarms, analyze logs, and troubleshoot performance issues using CloudWatch
DATE: 14-11-25	

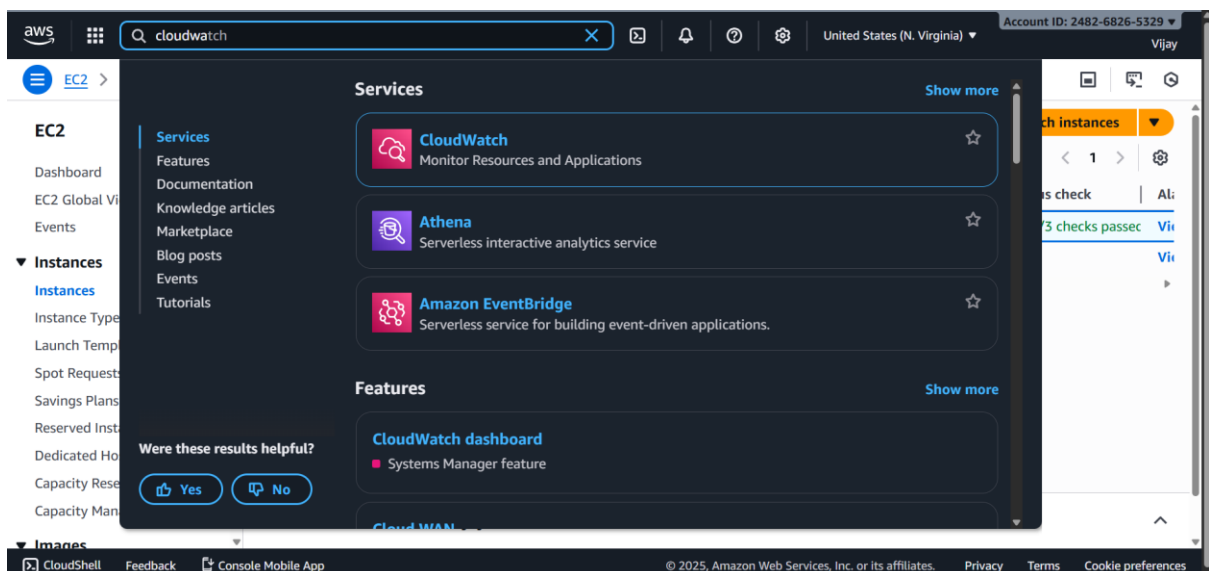
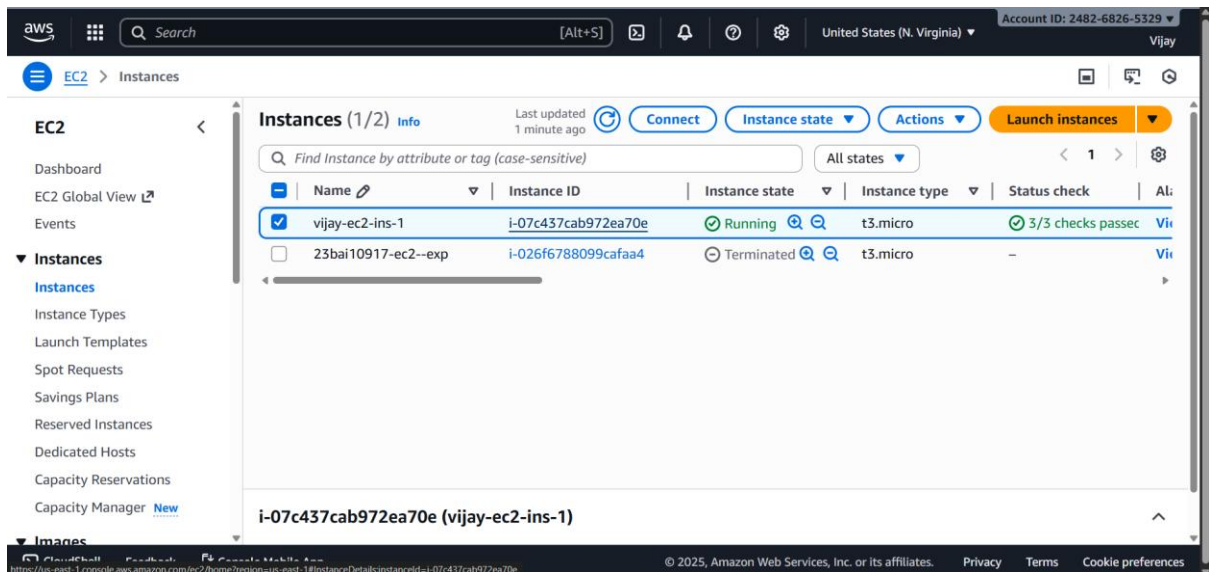
AIM

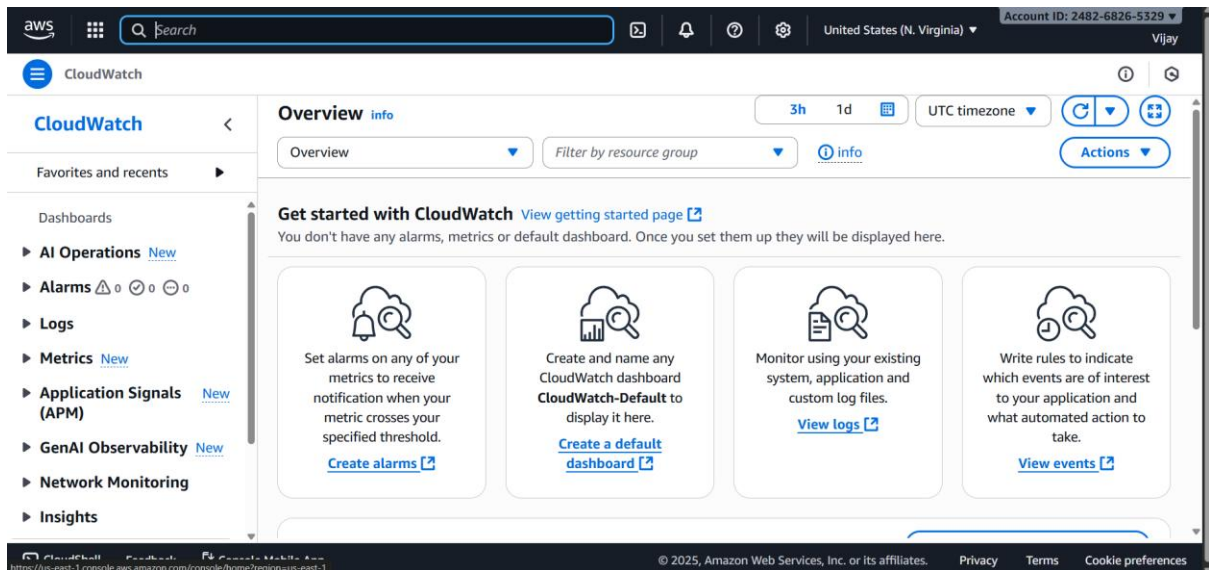
To Set up CloudWatch monitoring for AWS resources, create custom metrics and alarms, analyze logs, and troubleshoot performance issues using CloudWatch

PROCEDURE

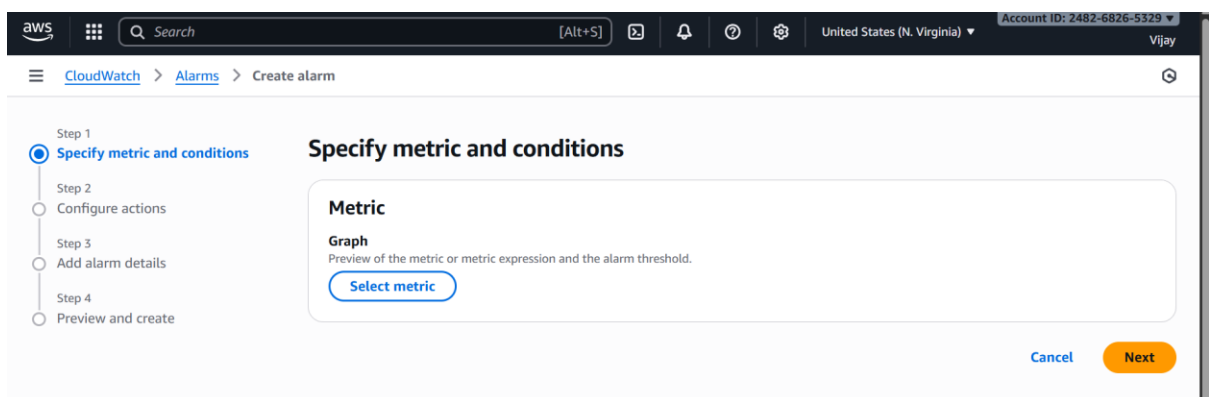
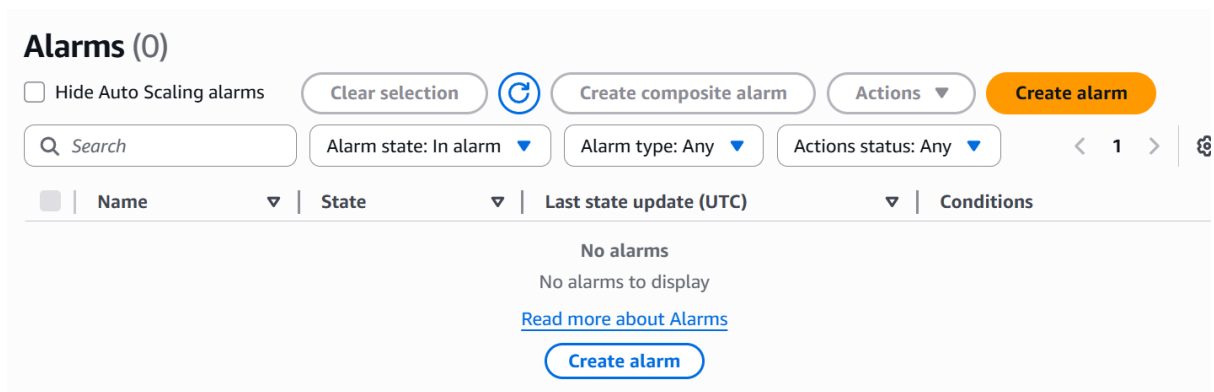
1. Provision and Instrument AWS Resource
2. Launch an EC2 instance (e.g., t2.micro) and ensure detailed monitoring is enabled. For metrics beyond the basic 5-minute interval (like memory usage or disk utilization), install and configure the CloudWatch Agent on the instance. The agent will also be used to collect log files.
3. Create Custom Metrics
4. Using either the CloudWatch Agent configuration (for system metrics like memory/disk) or by creating Metric Filters in CloudWatch Logs, define a custom metric. For a Metric Filter, search the logs for a specific pattern (e.g., "ERROR" or a specific response time) and assign it a metric value (e.g., count the occurrences of "ERROR").
5. Set Up Alarms and Notifications
6. Create a CloudWatch Alarm based on one of the collected metrics. Choose a standard metric (e.g., EC2 CPU Utilization) or your newly created custom metric. Define the threshold (e.g., CPU Utilization $> 70\%$ for 5 minutes). Configure an SNS Topic to send an email notification when the alarm state changes to ALARM.
7. Simulate Performance Issue
8. On the EC2 instance, execute a command or script (e.g., using the stress utility for Linux) to deliberately increase the monitored metric (e.g., 100% CPU utilization). This action is intended to breach the alarm threshold and generate logs.
9. Monitor the CloudWatch Metrics graph for the resource to observe the increase in the metric and verify that the alarm state transitions to ALARM. Confirm that the configured SNS notification is received. Stop the performance simulation to see the metric return to normal.
10. Navigate to CloudWatch Logs Insights. Select the log group configured in Step 2. Execute a query to filter and analyze the logs generated during the performance issue (e.g., search for errors, count requests, or calculate average latency). Use the log data to identify the root cause of the simulated performance issue.

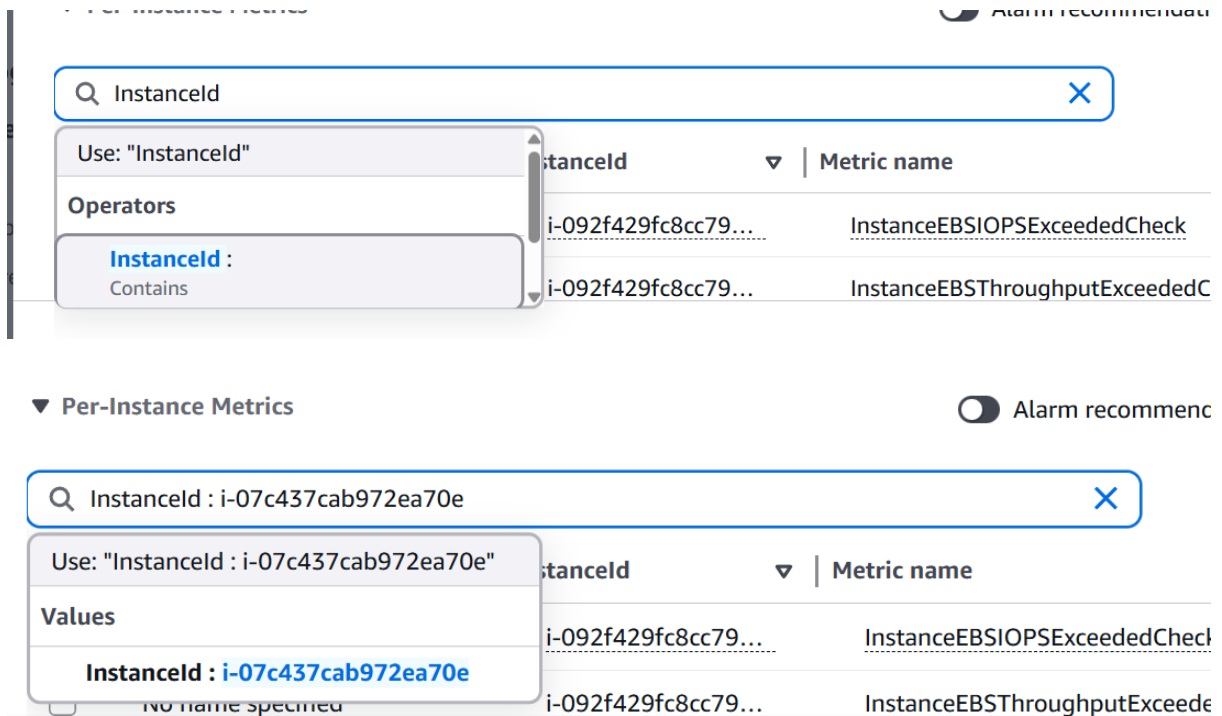
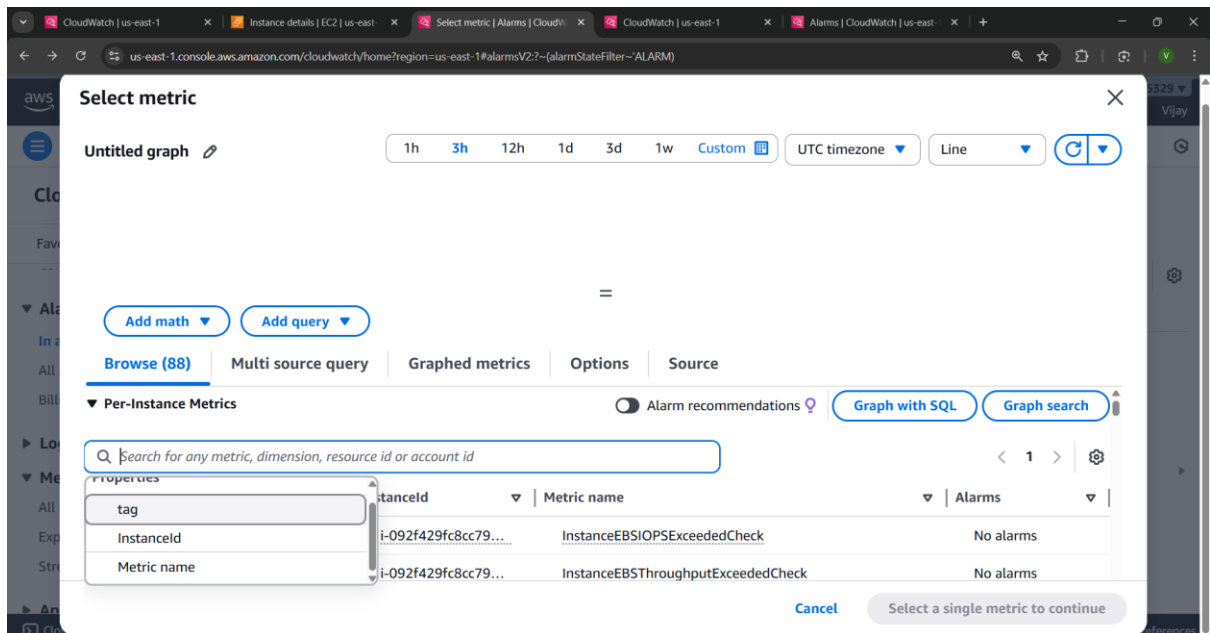
IMPLEMENTATION SCREENSHOT





Alarms & Metrics





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Account ID: 2482-6826-5329
Vijay

CloudWatch > Alarms > Create alarm

Period
5 minutes

Conditions

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...
Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
>= threshold

☐ Lower/Equal
<= threshold

☐ Lower
< threshold

than...
Define the threshold value.

Must be a number.

▶ Additional configuration

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Account ID: 2482-6826-5329
Vijay

CloudWatch > Alarms > Create alarm

1 Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 1
Specify metric and conditions

☒ Step 2
Configure actions

☐ Step 3
Add alarm details

☐ Step 4
Preview and create

Configure actions

Notification
Add notification

Lambda action
Add Lambda action

Auto Scaling action
Add Auto Scaling action

EC2 action

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United States (N. Virginia)
Account ID: 2482-6826-5329
Vijay

CloudWatch > Alarms > Create alarm

☒ Configure actions

☐ Step 3
Add alarm details

☐ Step 4
Preview and create

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ In alarm
The metric or expression is outside of the defined threshold.

☐ OK
The metric or expression is within the defined threshold.

☐ Insufficient data
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☐ Select an existing SNS topic

☒ Create new topic

☐ Use topic ARN to notify other accounts

Create a new topic...
The topic name must be unique.

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...
Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

user1@example.com, user2@example.com

Create topic

Add notification

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CloudWatch > Alarms > Create alarm

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Name and description

Alarm name

vijay-ec2-ins-alarm-1

Alarm description - optional [View formatting guidelines](#)

Edit

Preview

¶ This is an H1

****double asterisks will produce strong character****

This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional [info](#)

No tags associated with the resource.

Add new tag

You can add up to 50 tags

Cancel

Previous

Next

CloudWatch > Alarms > Create alarm

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Preview and create

Step 1: Specify metric and conditions

Metric

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent

90

45

0.089

02:00

02:30

03:00

03:30

04:00

04:30

CPUUtilization

Namespace

AWS/EC2

Metric name

CPUUtilization

InstanceId

i-07c437cab972ea70e

Instance name

vijay-ec2-ins-1

Statistic

Average

Period

5 minutes

Conditions

Threshold type

Cancel

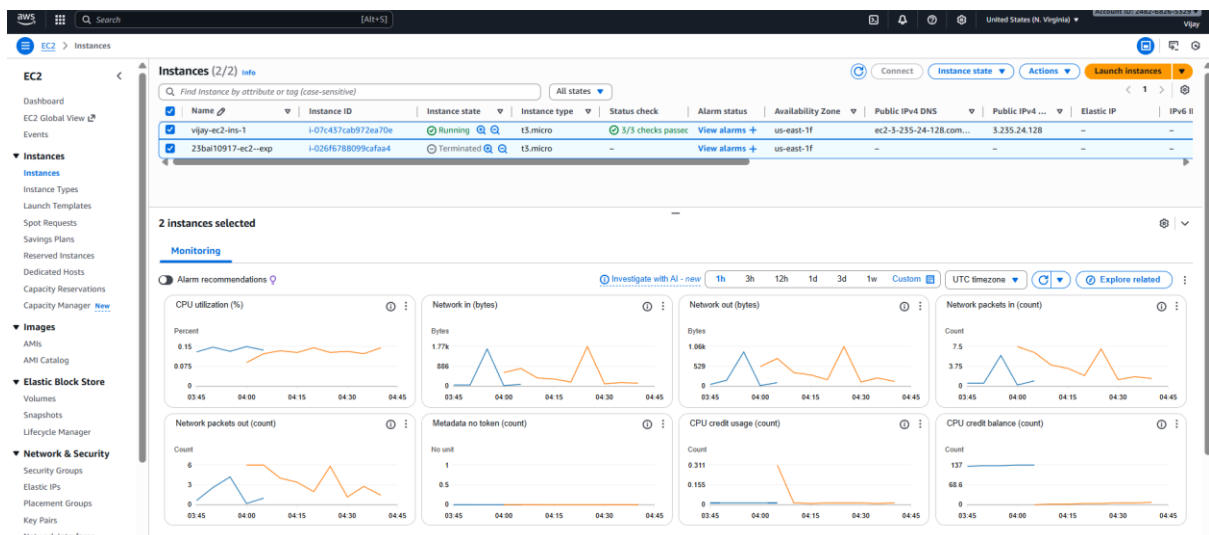
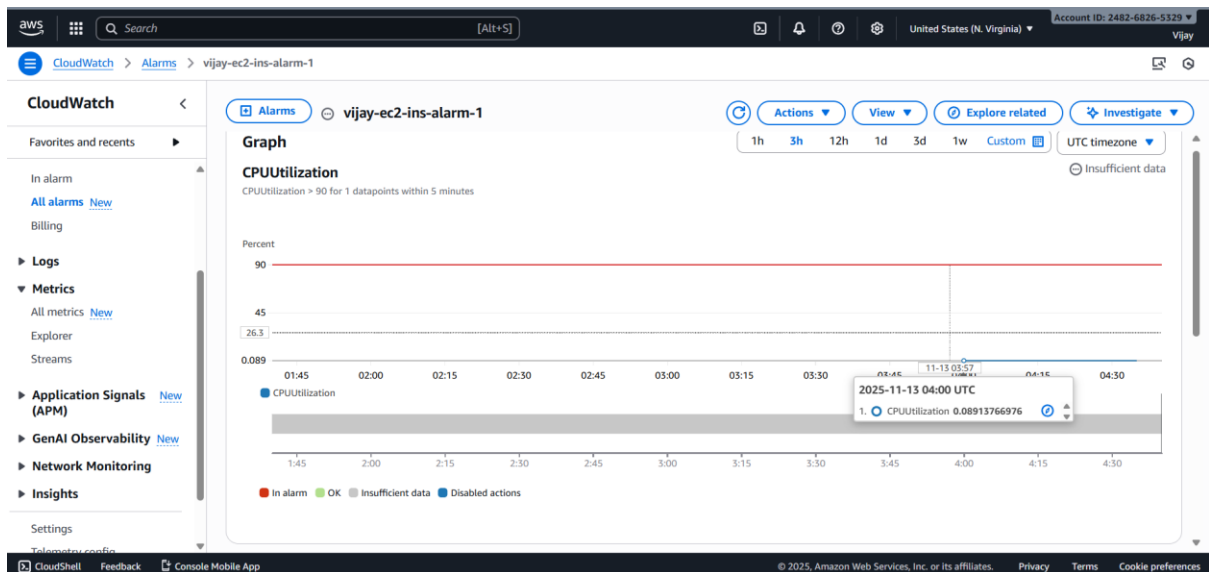
Previous

Create alarm

Successfully created alarm vijay-ec2-ins-alarm-1.

View alarm

47



RESULT

The successful execution of the CloudWatch lab demonstrated the ability to collect and visualize metrics (CPU, Custom Metrics), confirm alarm triggering via SNS notification upon threshold breach, and utilize CloudWatch Logs Insights for querying log data to effectively troubleshoot and identify the root cause of the simulated performance issue.